

Instituto de Matemáticas



The Mckay correspondence

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1. Introduction

Abstract Mathematics

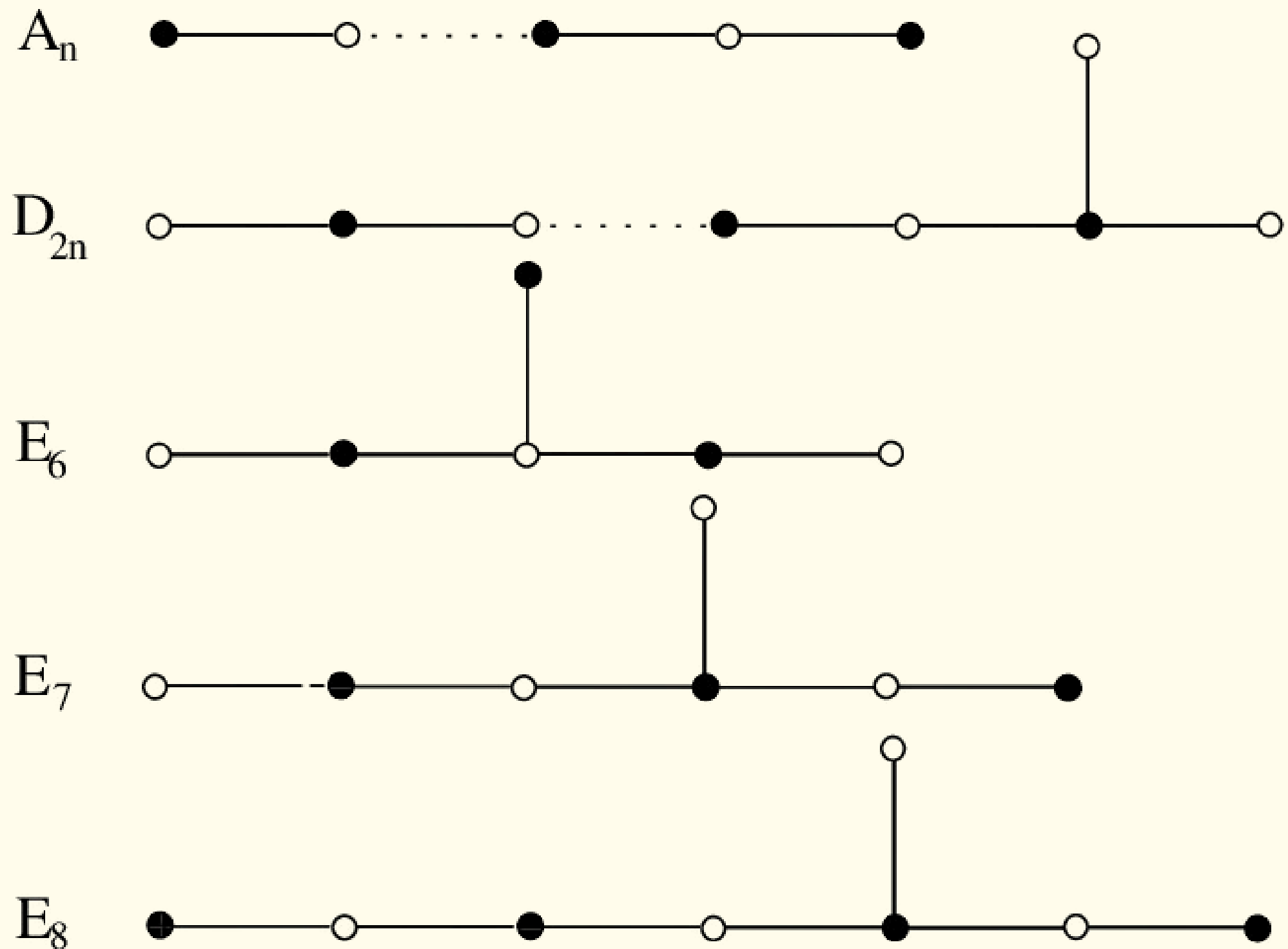
Mathematical structures
for understand Physics



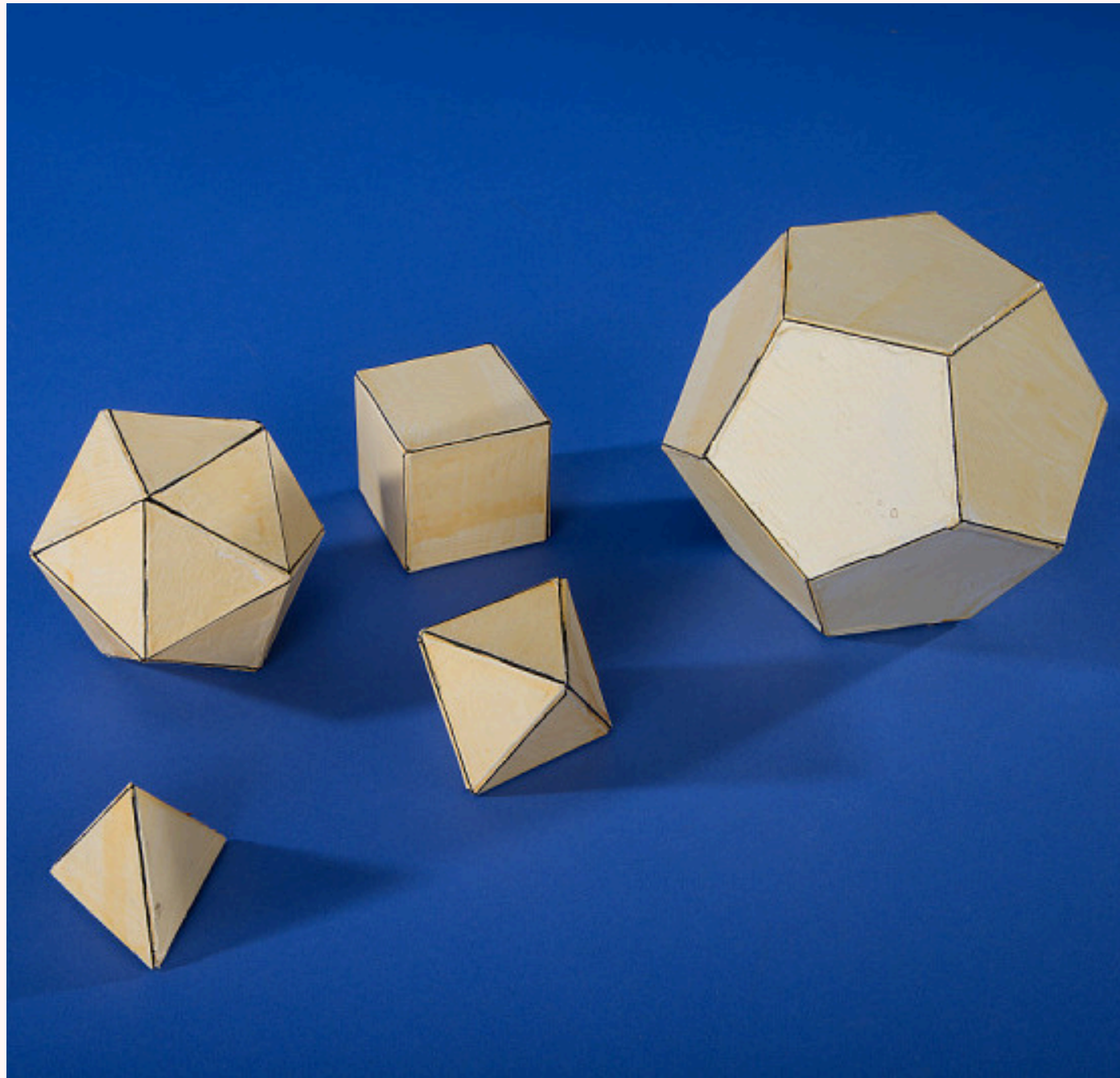
Connections

Easy codes or draws
for complex concepts

2. ADE Dynkin diagrams.

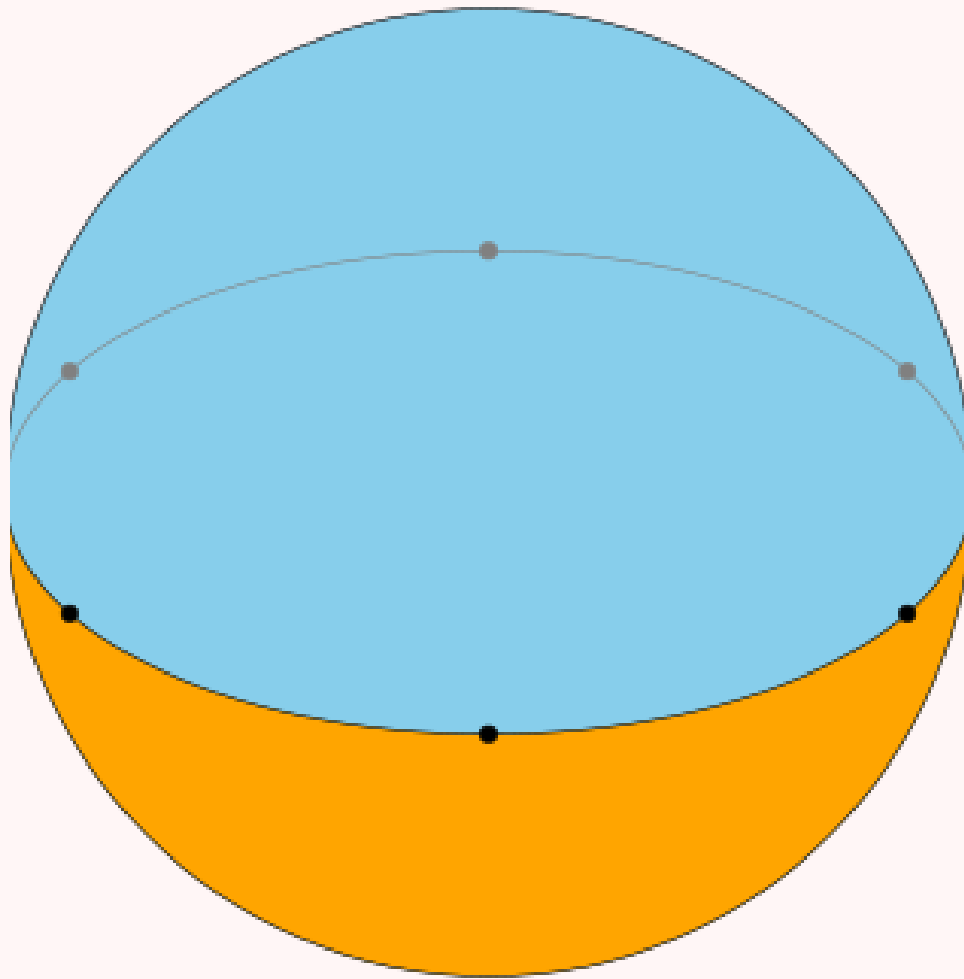


3. Groups of rotations.

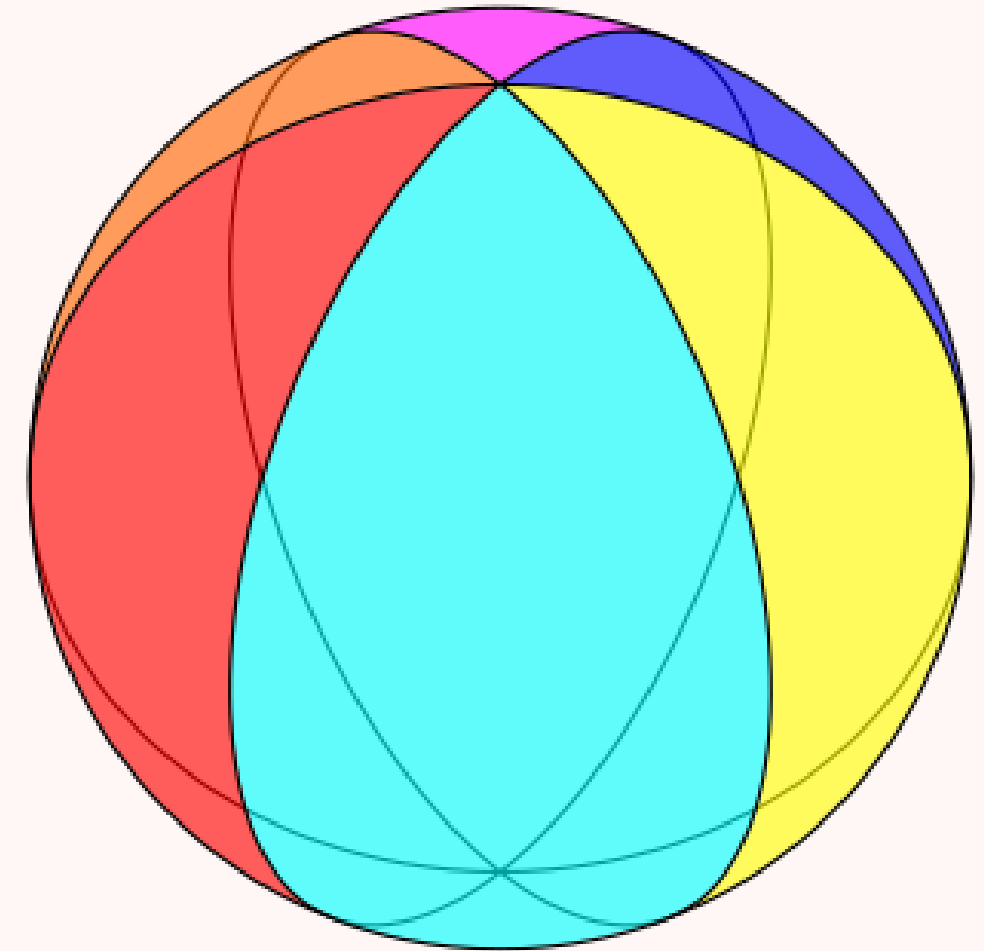


The **5** Platonic solids
(only **3** classes for us)

3. Groups of rotations.



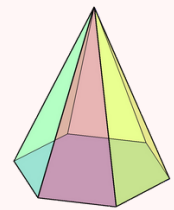
Dihedron



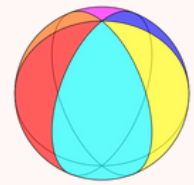
Hosohedron

3. Groups of rotations.

Rotations of



Regular pyramid



Dihedron (hosohedron)



Tetrahedron

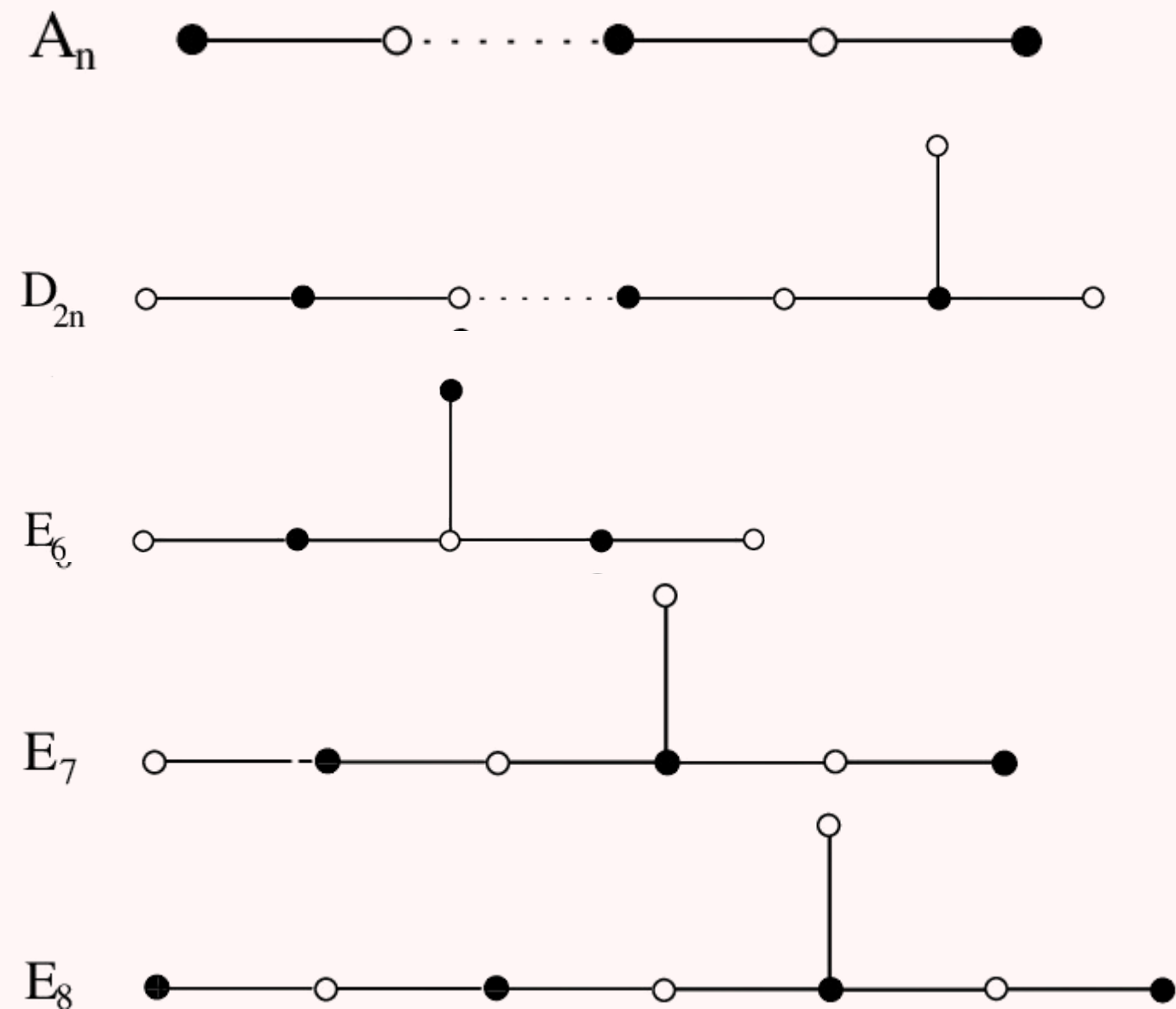


Cube (octahedron)

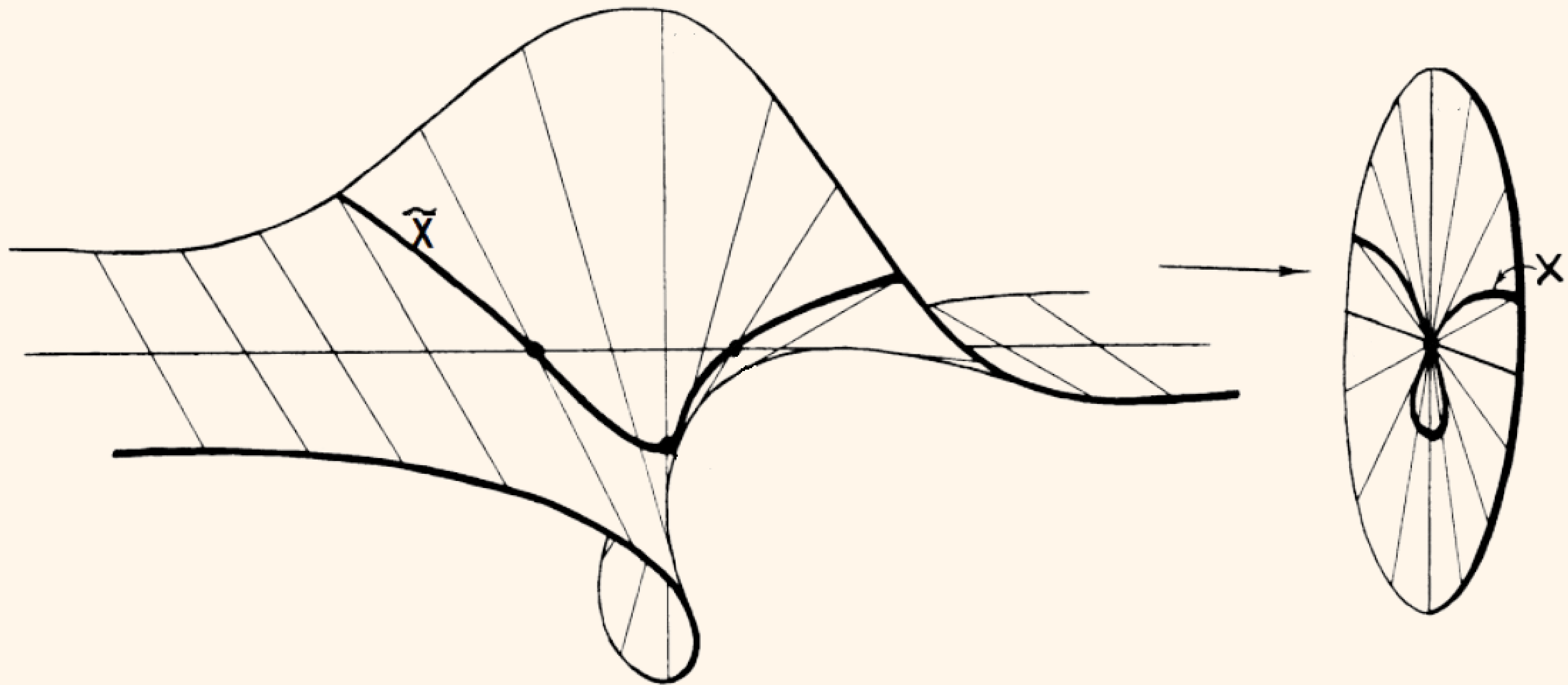


Dodecahedron
(icosahedron)

ADE Dynkin Diagram



4. Algebraic varieties.



4. Algebraic varieties.

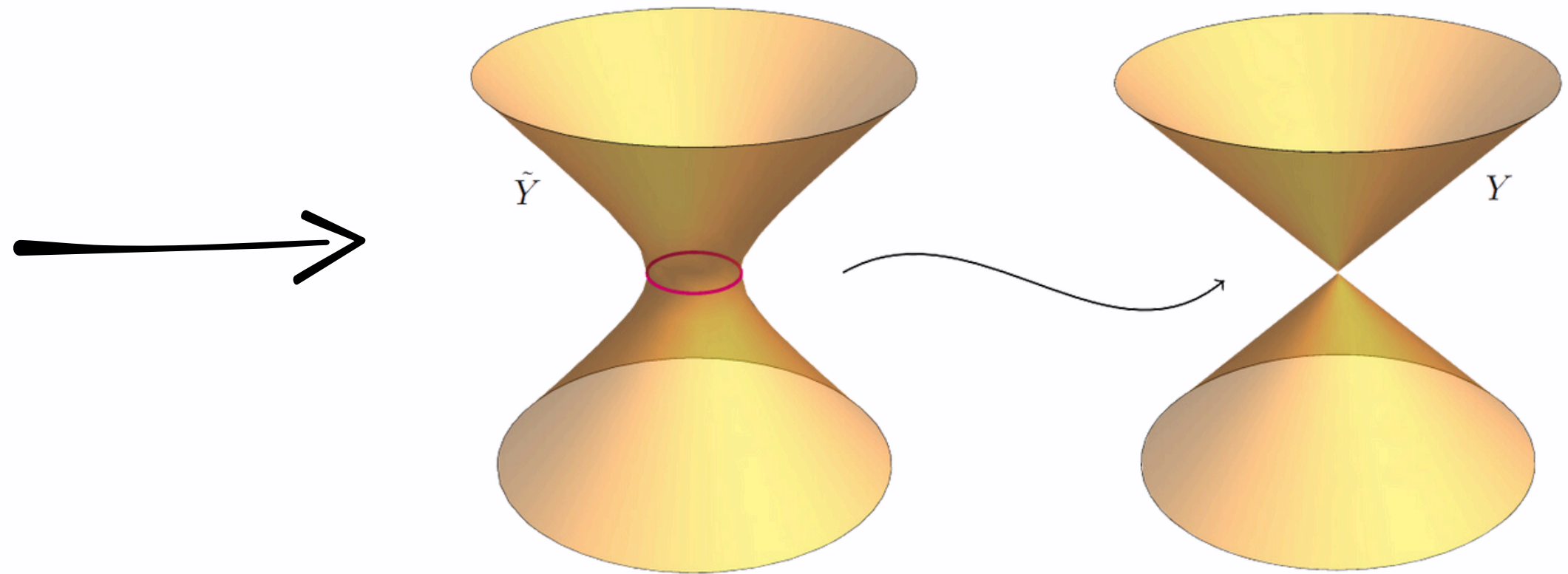
Group of rotations
in the space



An irregular
algebraic variety

4. Algebraic varieties. Example.

Group of **rotations** of
the **regular pyramid**



5. The McKay Correspondence.

Group of rotations
in the space



ADE Dynkin **diagrams**



Resolution of
the irregular
algebraic variety

1980



6. Principal Bibliography.

- John McKay, Graphs, singularities, and finite groups Proc. Symp. Pure Math. Vol. 37. No. 183. (1980)
- Gérard González-Sprinberg, Jean-Louis Verdier, Construction géométrique de la correspondance de McKay, Ann. Sci. École Norm. Sup. Vol. 16. No.3 (1983)

Q & A time.

Thank you for your attention!